

[OPEN SOURCE] Tactical Near-Field Medical Scanner via De-commissioned Military AESA Radars (Project: Glass Hunter)

[1.] "This is a theoretical conceptual study.

2.] "I do not have access to any classified military data."

3.] **"Implementation must be done by certified engineers and medical professionals."]**

"This Tech is for Humans, not for Politics. Build it if you want to save lives."

1. Hardware Requisition Targets (The "Donors")

Targeting surplus/decommissioned 4.5-gen fighter radar upgrades currently rotting in NATO warehouses.

* Primary Targets: Early-gen AN/APG-63(V)2 / (V)3 array faces (stripped from F-15C/Es).

* Secondary Targets: Legacy F-16 SABR modules or early Eurofighter Captor-E prototypes.

* Why: We only need the physical Array Face packed with thousands of GaAs (Gallium Arsenide) or GaN (Gallium Nitride) T/R (Transmit/Receive) modules. They are aerospace-grade, lightweight, and indestructible.

2. ITAR Bypass via "De-chipping" (Sanitization)

To bypass military security and export control concerns (e.g., ITAR), we physically lobotomize the radar.

* REMOVE (The Brains): Back-end Processors, Crypto Key Loaders, and target-identifying Military FPGAs. These hold the classified algorithms for missile guidance. Strip them and hand them back to the MoD/Pentagon.

* KEEP (The Muscle): The Antenna Array Face and the Power/Cooling Manifolds. Without the back-end processors, this is no longer a weapon—it is merely a high-performance, programmable RF emitter.

3. Detection Mechanism: Near-Field SAR & Dielectric Contrast

Repurposing X-band/Ka-band from tracking jets at 150km to tracking

shrapnel at 30cm.

* The Physics: Radar detection relies on the Dielectric Constant (ϵ). Human tissue (mostly water) has an $\epsilon \approx 80$, which absorbs or transmits the RF waves. Glass/silicon shrapnel has an $\epsilon \approx 4 \sim 10$, causing massive backscattering. The human body becomes transparent; the glass lights up like a beacon.

* Resolution Formula: By utilizing the massive bandwidth (B) of the AESA in a Near-Field Synthetic Aperture Radar (SAR) configuration, we achieve extreme spatial resolution (Δx):

(Where c is the speed of light. Maxing out the bandwidth yields sub-millimeter resolution, creating a perfect 3D coordinate map of the glass shards).

* COTS Brains: Wire the stripped array to COTS (Commercial Off-The-Shelf) GPUs (e.g., Nvidia RTX series) or a Raspberry Pi cluster. Load an open-source near-field imaging algorithm to render the coordinates.

4. Field Expediency: Low-Power Mode & Survivability

Designed for brutal frontline triage units, not pristine hospitals.

* 220V AC Plug-and-Play (Power Scaling): * Military AESAs draw tens of kW to see across borders. We only need to see 1 meter deep. We throttle the Tx power down to the mW (milliwatt) range.

* By attaching a standard 220V AC to DC SMPS (Switched-Mode Power Supply), the system can run on a standard wall outlet or an ambulance battery. The radiation output is on par with a commercial Wi-Fi router. 100% safe for prolonged medical use.

* “Pong-Pong” Survivability (Conformal Coating):

* Field medics don’t have time for sterile wipes. They will use standard dish soap (surfactants) and water to scrub blood off the equipment.

* Action: Apply 2–3 layers of commercial Urethane or Silicone Conformal Coating directly over the array face. This waterproofs the T/R modules and prevents electrical shorts without degrading RF transparency. Scrub it with a sponge; it won’t die.

[Architect’s Note]

“I’ve laid out the physics, the ITAR bypass, and the tactical field modifications. The components are currently collecting dust in your warehouses. The coding takes a week for any decent engineering team. Do not wait for bureaucracy. Rip the arrays, spray the coating, plug it into the wall, and start pulling glass out of patients. The blueprint is yours. The action is on you.”

[Architect's Final Statement]

"I didn't write this for any specific nation, flag, or political agenda. I wrote this because I am tired of seeing children and non-combatants suffer from invisible wounds that modern field medicine fails to see.

This is a gift from a free engineer to those who value human life over scrap metal. If you have these arrays rotting in your hangars, you have the cure for a thousand agonizing surgeries in your hands. I have given you the map; whether you choose to walk the path or let the dust collect is on your conscience.

This tech is for the humans. Period."

Branzino

[**"Note on Hardware Models:** Specific military designations (e.g., AN/APG-63, SABR) are mentioned solely as **conceptual benchmarks** for required technical specifications (bandwidth, power, and array density). I do not advocate for the unauthorized acquisition of active military assets. These are examples of legacy technology that should be legally repurposed for humanitarian aid."]